Responses to Reviewers' Comments for Manuscript PR-D-26-10575

# LLM-Driven Unified Framework with Grid Diffusion for Efficient Long Video Editing

Addressed Comments for Publication to

Pattern Recognition

by

Deyin Liu, Li He, Zhan Li, Chengwu Liang

Dear Petia I. Radeva, Dr.,

Please find enclosed the revised version of our previous submission entitled “LLM-Driven Unified Framework with Grid Diffusion for Efficient Long Video Editing” with manuscript number PR-D-26-10575. We would like to thank you and the reviewers for the valuable comments which help improve the quality of our manuscript significantly.

Sincerely,

Prof. Zhe Jin
Anhui University

**Note:** To enhance the legibility of this response letter, all the editor’s and reviewers’ comments are typeset in boxes. Rephrased or added sentences are typeset in color. The respective parts in the manuscript are highlighted to indicate changes.

# Response to the Meta Review

**Summary Comment 1**

The submission has been substantially by a range of reviews across multiple respects. The reviewers recognise the proposed framework promising, but its central claims are insufficiently validated. Major revision is required to quantitatively evaluate the LLM router, grid/sliding ablations, long-term temporal consistency, and end-to-end efficiency, while clarifying metric definitions, implementation details, and the scope of long-video claims.

**Response S1:** We highly value the meta-review and the comprehensive assessments from all reviewers. We have thoroughly examined the entire manuscript and made targeted revisions to resolve every raised concern. In what follows, we list the meta-review and reviewer comments, alongside our itemized responses to address all issues.

# Response to Reviewer 1

Weaknesses:

**Comment 1.1**

None of the three claimed mechanisms has a quantitative ablation. Grid sliding is validated on one video by eye (Fig. 7), grid size on one lion video with the outcome described as "overall beauty is improved but the details are coarse", and the LLM on three videos (Fig. 9) — all qualitative. For a paper whose contributions are precisely these three mechanisms, this is not sufficient evidence. Suggestion: add a table over the full evaluation set reporting all four metrics for abaltion experiments.

**Response 1.1:**

We appreciate this critical comment and fully agree that quantitative ablations are essential to verify our three core mechanisms. Ablation experiments across the evaluation dataset have been supplemented in the revised manuscript. A complete ablation table is provided to report all four quantitative metrics for each variant, covering grid sliding, grid size, and the LLM router module.

Table 1: Quantitative ablations covering grid sliding, grid size, and the LLM router module. We adopt the attention control scheme as our image editing approach. Since the grid size ablation experiments require fixed frame settings, their metric values fluctuate compared with those of the other two ablation experiments.

| Setting | Masked PSNR ↑ | SSIM ↑ | CLIP-T ↑ | Tem-Con ↑ |
|---|---|---|---|---|
| w (grid sliding) | 22.71 | 0.6424 | 0.2936 | 0.9619 |
| w/o (grid sliding) | 22.67 | 0.6412 | 0.2935 | 0.9311 |
| $1 \times 1$ (grid size) | 21.91 | 0.6220 | 0.2942 | 0.9205 |
| $2 \times 2$ (grid size) | 21.86 | 0.6221 | 0.2937 | 0.9638 |
| $3 \times 3$ (grid size) | 21.83 | 0.6214 | 0.2926 | 0.9651 |

**Comment 1.2**

The LLM router is never isolated, and Figure 9(a) appears to contradict its operating principle. Settings 1–3 differ in configuration, not in the presence of a router, so the experiment shows only that fixed configurations are inadequate; the LLM's own accuracy, its value over a keyword rule, its latency and token cost (absent from Table 2), and its sensitivity across LLMs are all unmeasured. Meanwhile Fig. 9(a) lists identical source and target prompts ("a girl is dancing" -> "a girl is dancing") for a case the router is said to classify as motion editing with fine-tuning — impossible for a text-only prompt diff. Suggestion: report routing accuracy against human labels on > 100 prompt pairs, with an oracle upper bound and a rule-based lower bound, plus router latency; and correct or explain Fig. 9(a).

**Response 1.2:**

We greatly appreciate your valuable comments. We explain the experimental setup in Figure 9. Setting 3 utilizes an LLM for adaptive model adjustment according to text prompts.

Setting 1 removes the masking mechanism without fine-tuning, and Setting 2 adopts a fixed masking mechanism and fine-tunes temporal attention. Setting 1 and Setting 2 adopt fixed model configurations, where the LLM router is removed, which corresponds to the case of isolating the LLM router. By comparing the performance of Setting 1, Setting 2 and Setting 3, we can obtain the performance gain brought by our LLM module. To intuitively demonstrate the benefits of the LLM module, we have supplemented quantitative ablation results of the LLM module, as shown in Table 1. Due to the lack of editing masks in Setting 1, editing leakage occurs. Its Masked PSNR, SSIM and Tem-Con metrics are poor, while the CLIP-T metric achieves a relatively high value because there is no mask constraint. Setting 2 fine-tunes the temporal module and achieves decent performance across all metrics, but it consumes substantial unnecessary time and computational resources for fine-tuning.

We have supplemented the confusion matrix to evaluate the accuracy of the LLM router, as shown in Table 2. The Average precision of the LLM router exceed 93 percent, which verifies its reliability. We use DeepSeek-v4 as our default LLM. Its average single inference latency is 0.51 seconds, with an average token cost of 1934 tokens. In addition, we test the robustness of our LLM router across different LLMs, and the evaluation results are shown in Table 3. There are slight performance gaps among different LLMs, which stem from the inherent capabilities of each LLM. We select DeepSeek-v4 after comprehensively considering performance, cost and usability. Readers can freely switch to more powerful LLMs to build the LLM router for higher routing accuracy.

Figure 9(a) shows the motion editing results under Setting 3. Our motion editing module is adapted from MotionEditor, which requires an additional motion guidance map to guide the generation of new motions. To enable the LLM to correctly recognize this as a motion editing task, our code appends the filename of the motion guidance map to the target prompt as a hint for the LLM. Therefore, the actual prompt pair is ‘a girl is dancing” → ‘a girl is dancing (/motion guidance filename)”. This is merely an engineering trick in our implementation. We omitted this detail in the figure, which caused misunderstanding. Thanks for your careful observation; we will correct Figure 9(a).

Table 2: Confusion matrix of editing type prediction.

| Prediction \ Ground Truth | Local | Global | Simple | Complex | Motion | Total | Precision |
|---|---|---|---|---|---|---|---|
| Local Editing (need mask) | 98 | 4 | — | — | — | 102 | 0.9608 |
| Global Editing (no mask) | 2 | 96 | — | — | — | 98 | 0.9796 |
| Simple Editing (no fine-tune) | — | — | 91 | 6 | 5 | 102 | 0.8922 |
| Complex Editing (need fine-tune) | — | — | 7 | 89 | 0 | 96 | 0.9271 |
| Motion Editing (need motion tune) | — | — | 2 | 5 | 95 | 102 | 0.9314 |
| Total/average | 100 | 100 | 100 | 100 | 100 | 500 | 0.9382 |

Questions for authors:

**Comment 1.3**

What are UELVE-Attention’s four metrics with the FastSAM mask disabled? Since Masked PSNR rewards background copying and UELVE is the only masked method in Table 1, this is needed to attribute the Masked PSNR win to the grid mechanism.

Table 3: Robustness experiments on different LLMs.

| LLM | Local-F1 | Global-F1 | Simple-F1 | Complex-F1 | Motion-F1 | Macro-F1 |
|---|---|---|---|---|---|---|
| DeepSeek v4 | 0.9703 | 0.9697 | 0.9010 | 0.9082 | 0.9406 | 0.9380 |
| GLM 5.2 | 0.9490 | 0.9770 | 0.9090 | 0.9160 | 0.9480 | 0.9400 |
| Qwen-2 | 0.9430 | 0.9785 | 0.8880 | 0.8960 | 0.9420 | 0.9295 |
| Llama3 | 0.9310 | 0.9540 | 0.8760 | 0.8980 | 0.9435 | 0.9205 |
| doubao-seed-2.0-lite | 0.9160 | 0.9555 | 0.8600 | 0.8995 | 0.9070 | 0.9076 |
| kimi-k2.6 | 0.9175 | 0.9380 | 0.8490 | 0.9010 | 0.9085 | 0.9028 |
| oracle (human) | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |

**Response 1.3:**

Thanks for this valuable comment, which helps us clarify the source of performance gain on Masked PSNR.

UELVE is not the only mask enabled method in Table 1. For instance, Tune-A-Video, Video-P2P and FateZero all employ masking mechanisms. Our Masked PSNR is comparable to these mask based approaches. The marginal performance advantage originates from the higher mask precision of FastSAM and inherent limitations of those baselines, rather than our grid mechanism. Tune-A-Video relies solely on the structural information preserved during inversion to guide editing, which leads to severe corruption of the background. The fine-tuning and null-text optimization in Video-P2P partially distort the original video tone in terms of brightness and contrast. As for FateZero, it reshapes the original self-attention into spatial-temporal self-attention, which induces background distortion and degrades Masked PSNR.

The core emphasis of UELVE is to flexibly decide whether to enable masking via prompt semantic analysis. This design reduces unnecessary mask generation overhead and avoids the inconvenience brought by manually controlled mask parameters, instead of pursuing higher mask quality. The specific choice of masking scheme is not critical to our method. Replacing FastSAM with alternative mask generation approaches of equivalent performance will not yield obvious metric variations. Naturally, adopting a lower quality masking mechanism will inevitably hurt Masked PSNR. As shown in Table 4, we disable FastSAM and adopt the same masking mechanism used in other baselines. After this modification, UELVE achieves reduced Masked PSNR and SSIM, a slight improvement in CLIP-T, and nearly unchanged Tem-Con.

We acknowledge that our original use of a different masking mechanism compared with baselines indeed creates an unfair comparison. Thank you for pointing this out, and we will revise accordingly.

Table 4: We disable FastSAM and adopt the same masking mechanism used in other baseline methods. After this modification, UELVE achieves reduced Masked PSNR and SSIM, a slight improvement in CLIP-T, and nearly unchanged Tem-Con.

| **Method** | **Masked PSNR↑** | | | | **SSIM↑** | | | | **Clip-T↑** | | | | **Tem-Con↑** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Tune-A-Video [**TAV**] | 13.91 | 15.87 | 15.13 | — | 0.4350 | 0.4749 | 0.4418 | — | 0.2960 | 0.2727 | 0.2894 | — | 0.9488 | 0.9481 | 0.9441 | — |
| Video-P2P [**Video-P2P**] | 23.66 | 22.23 | — | — | 0.6563 | 0.6217 | — | — | 0.2937 | 0.2955 | — | — | 0.9695 | 0.9640 | — | — |
| FateZero [**Fatezero**] | 23.83 | — | — | — | 0.6572 | — | — | — | 0.2872 | — | — | — | 0.9576 | — | — | — |
| UELVE (Cross-Attn) | 23.86 | 22.66 | 21.64 | 21.57 | 0.6840 | 0.6627 | 0.6168 | 0.5827 | 0.2847 | 0.2962 | 0.2926 | 0.2898 | 0.9655 | 0.9654 | 0.9594 | 0.9725 |
| UELVE (FastSAM) | 24.05 | 22.83 | 21.78 | 21.65 | 0.6849 | 0.6635 | 0.6176 | 0.5833 | 0.2841 | 0.2955 | 0.2922 | 0.2893 | 0.9653 | 0.9655 | 0.9592 | 0.9725 |

**Comment 1.4**

How many MB are exchanged per timestep per boundary at 200 frames with G = 2, what fraction of step time is communication, and how does the exchange scale for G=4,8? Relatedly, does edited-video quality (not just speed) degrade as G increases, given that more GPUs means more boundaries over the same 50 steps?

**Response 1.4:**

Data exchange only occurs at the grid boundaries between adjacent GPUs. For an original video resolution of 512×512 and a 2×2 grid layout, when G=2 (GPUs) with a total of 200 frames, latent data of 2 frames is transmitted per inference timestep, which is approximately 0.0625 MB. The communication latency is around 30 ms, accounting for less than 1 percent of the total per-timestep runtime. The communication overhead rises to 90 ms for G=4 and 210 ms for G=8. In general, for G=n, the communication latency equals $(n-1) \times 30$ ms. Multi-GPU execution only distributes computational workload across devices and does not alter the inference pipeline. The model operates identically under single-GPU and multi-GPU settings, so the edited video quality does not degrade as the number of GPUs increases.

**Comment 1.5**

How many MB are exchanged per timestep per boundary at 200 frames with G = 2, what fraction of step time is communication, and how does the exchange scale for G=4,8? Relatedly, does edited-video quality (not just speed) degrade as G increases, given that more GPUs means more boundaries over the same 50 steps?

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**Comment 1.6**

FastSAM is class-agnostic. Which component maps the changed noun phrase to a specific segment, and how is that segment propagated across frames, across grids, and across the two alternating layouts?

**Response 1.6:**

FastSAM is indeed a class-agnostic segmentation model. We leverage the CLIP model to map the modified noun phrase to the target segmented region. We generate masks for each individual frame, and these masks are bound to frame indices. Therefore, mask propagation across frames is unnecessary. Similarly, no propagation is required across grids or between the two alternating layouts.

# Response to Reviewer 2

Reviewer 2: The 2x2 grid and alternating layouts are sensible engineering choices, and the qualitative results suggest that the approach can be useful. The unresolved issue is the distance between the paper's broad claims and what is actually measured. In particular, the reliability of the LLM router, the cost of processing genuinely long clips, and the fairness of the timing comparison are not yet clear.

**Comment 2.1**

Clip-T needs correction. Section 4.1.4 calls Clip-T an average CLIP distance, but Table 1 marks it with an upward arrow. If it is a distance, lower should normally be better; if it is a similarity, the formula and description need to say so. Give the exact definition and correct the table and discussion before drawing conclusions from this metric.

**Response 2.1:**

Thank you very much for your careful review and valuable comments, which greatly help us improve our paper. The CLIP-T evaluation metric adopted in our work is referenced from the RAVE method. The original description in the RAVE paper states that "Additionally, we evaluate the textual alignment of the editing, given a prompt, by calculating the average distance between the CLIP embedding of the text prompt and all frames (CLIP-T)", where it is defined as cosine distance. However, after verifying the official code of RAVE, we found that the metric is actually implemented as cosine similarity, which is a writing error in the original RAVE paper. We will correct the description in our manuscript and uniformly define CLIP-T as cosine similarity in the revised version.

The precise definition is:

$$\text{CLIP-T} = \frac{1}{T}\sum_{t=1}^{T}\frac{\langle f_{\text{text}}, f_{\text{img}}^{(t)}\rangle}{\|f_{\text{text}}\|_2\,\|f_{\text{img}}^{(t)}\|_2},$$

where $f_{\text{text}}$ is the CLIP text embedding of the prompt, $f_{\text{img}}^{(t)}$ is the CLIP image embedding of the $t$-th frame, and $T$ is the number of frames. CLIP-T is a similarity, so higher is better.

**Comment 2.2**

In the present system, DeepSeek seems mainly to choose among existing options such as masking, temporal fine-tuning, Attention Control, and ControlNet. The paper does not give the model version, prompt, decoding settings, or error handling. Fig. 9 shows three examples, but it does not measure how reliably the router makes the right choice. A small quantitative comparison with a fixed setting and a rule-based or LLM-free router would be much more convincing. The same experiment should separate the gain from the LLM from the gain due to grid sliding and the existing editing modules.

**Response 2.2:**

**Model version, prompt, decoding settings, and error handling.** The LLM used in our router is DeepSeek-V4. The complete prompt is provided in the supplementary material. The decoding settings are temperature = 0.2 and max_tokens = 2048 with streaming output enabled (`stream=True`). For error handling, we adopt a retry mechanism: if the model output

cannot be parsed after the maximum number of retries, the system falls back to a conservative default configuration.

**Router reliability and ablation.** We conducted a quantitative evaluation of the routing accuracy of the LLM module. The confusion matrix and the corresponding evaluation metrics are reported in Table 2; the average precision metric exceeds 93%, indicating that the LLM router is reliable. Furthermore, as shown in Table 1, we performed a quantitative ablation study to separate the gain brought by the LLM from the gains due to grid sliding and the existing editing modules.

**Comment 2.3**

3.Eqs. (7)-(8) do not make the indexing, the symbol P, or the treatment of a video whose length is not divisible by the grid size sufficiently clear. State what happens to padding and boundary frames, and show one concrete example. A short pseudocode block would help more than another conceptual diagram. Quantitative results for 1x1, 2x2, and 3x3 grids would also support the choice of the default 2x2 layout.

**Response 2.3:**

Let $N = n \cdot l$ be the grid capacity. For a video of $K$ frames, we append $M - K$ copies of the last frame, where $M = \lceil K/N \rceil \cdot N$, so that $M$ is a multiple of $N$.

$$\text{Layout}_1 = \left\{ \mathcal{I}_1^{1,\ldots,N}, \ldots, \mathcal{I}_L^{P,\ldots,K,\ldots,M} \right\} \tag{1}$$

In Eq. (7), $P = (L-1) \cdot N + 1$ (equivalently $P = M - N + 1$) denotes the first frame index of the last grid, which may contain fewer than $N$ real frames and completed by the appended copies; the superscript of each $\mathcal{I}$ denotes the frame-index range it covers.

$$\text{Layout}_2 = \left\{ \mathcal{I}_0^{1,\ldots,n}, \mathcal{I}_1^{n+1,\ldots,N+n}, \ldots, \mathcal{I}_L^{P+n,\ldots,M} \right\} \tag{2}$$

In Eq. (8), sliding shifts every grid boundary forward by $n$ frames, which introduces one extra grid $\mathcal{I}_0^{1,\ldots,n}$. The appended copies frames participate only as attention context and are truncated before decoding, so the edited video has exactly $K$ frames.

Concrete example ($K = 13$, $2 \times 2$ grid: $n = 2$, $N = 4$, $M = 16$, $P = 13$; frames 14–16 are copies of frame 13):

Even steps ($\text{Layout}_1$): $\mathcal{I}_1^{1,2,3,4}$, $\mathcal{I}_2^{5,6,7,8}$, $\mathcal{I}_3^{9,10,11,12}$, $\mathcal{I}_4^{13,14,15,16}$ — four full 1024×1024 grids;

Odd steps ($\text{Layout}_2$): $\mathcal{I}_0^{1,2}$ (processed as a 512×1024 grid), $\mathcal{I}_1^{3,4,5,6}$, $\mathcal{I}_2^{7,8,9,10}$, $\mathcal{I}_3^{11,12,13,\underline{14}}$, $\mathcal{I}_4^{\underline{15,16}}$. Frames 14–16 are dropped before decoding; the output has exactly 13 frames.

As shown in Table 1, we conduct quantitative ablation experiments on the grid size. When the grid size is $1 \times 1$, frame consistency cannot be maintained via grid sliding, resulting in poor editing quality with a low Tem-Con metric. For a grid size of $3 \times 3$, attention calculations need to be performed on 9 frames within each grid simultaneously. Although the overall performance improves, the model loses fine-grained details and tends to produce blurriness and artifacts, which is reflected in the Masked PSNR, SSIM and CLIP-T metrics. Furthermore, the primary computational bottleneck of the transformer block lies in self-attention, whose computational cost scales quadratically with the image area. For the original video with 36 frames, using a $2 \times 2$ grid size means each grid contains 4 frames and the whole sequence is processed in 9 iterations. The total computational cost is approximately $4 \times 4 \times 9 = 144$ units.

**Algorithm 1** Grid diffusion with layout alternation

**Require:** frame latents $z_{1:K}$, grid shape $n \times l$, timesteps

$N \leftarrow n \cdot l$; $L \leftarrow \lceil K/N \rceil$; $M \leftarrow L \cdot N$; $P \leftarrow M - N + 1$

$z \leftarrow z_{1:K} \,\|\, \underbrace{z_K \cdots z_K}_{M-K}$ ▷ tail padding with copies of the last frame

**for** *step*, $t$ **in** *timesteps* **do**

  **if** *step* **is odd then**

    $grids \leftarrow \{\mathcal{Z}_0^{1,\ldots,n}, \mathcal{Z}_1^{n+1,\ldots,N+n}, \ldots, \mathcal{Z}_L^{P+n,\ldots,M}\}$ ▷ Layout$_2$, Eq. (2)

  **else**

    $grids \leftarrow \left\{\mathcal{Z}_1^{1,\ldots,N}, \ldots, \mathcal{Z}_L^{P,\ldots,M}\right\}$ ▷ Layout$_1$, Eq. (1)

  **end if**

  **for** each $\mathcal{Z}^{s,\ldots,e} \in grids$ **do**

    $\epsilon \leftarrow \mathrm{UNet}\big(\mathrm{grid}(z_{s:e}),\ t,\ \mathrm{prompt}\big)$

    $z_{s:e} \leftarrow \mathrm{DDIMStep}(\epsilon,\ t,\ z_{s:e})$

  **end for**

**end for**

**return** $\mathrm{Decode}\big(\mathrm{Restore}(z_{1:K})\big)$ ▷ drop appended copies; output has $K$ frames

When adopting a $3 \times 3$ grid size, each grid covers 9 frames and the sequence is processed in 4 iterations, yielding a total computational cost of roughly $9 \times 9 \times 4 = 324$ units, which imposes substantially higher computation requirements. Therefore, we select the $2 \times 2$ grid size as the default setting after comprehensive trade-offs.

as shown in Table 1, we performed a quantitative ablation study to

**Comment 2.4**

Table 2 mixes ordinary single-GPU baselines with RAVE x2GPU and UELVE x2GPU. It also reports inversion plus denoising, while UELVE may additionally spend time on LLM routing, segmentation, disk transfer, communication, and 300 steps of one-shot fine-tuning. Report the GPU count and peak memory for every row, and give an end-to-end time with and without the optional fine-tuning. Otherwise the claimed advantage over one-shot methods is difficult to interpret.

**Response 2.4:**

**Comment 2.5**

The paper says that arbitrary-length videos can be handled, but the quantitative results stop at 200 frames. The described one- to three-minute clips are much longer than this at the stated frame rates. Say whether full clips or selected windows were used. If full-length timing and quality results are not added, replace the broad claim with a more limited statement about the tested lengths and discuss disk-I/O and accumulated quality drift.

**Response 2.5:**

Table 5:

| **Method** | **GPU** | **8f** | | **16f** | | **48f** | | **200f** | |
|---|---|---|---|---|---|---|---|---|---|
| | | GMem | Time | GMem | Time | GMem | Time | GMem | Time |
| Tune-A-Video | 1 | 7.24GB | 5m03s | 10.27GB | 8m56s | 20.1GB | 27m17s | — | — |
| Video-P2P | 1 | 11.7GB | 5m25s | 15.8GB | 9m55s | — | — | — | — |
| FateZero | 1 | 13.9GB | 2m44s | — | — | — | — | —- | — |
| RAVE | 1 | 8.7GB | 1m48s | 10.2GB | 3m08s | 11.7GB | 8m37s | 12.4GB | 35m05s |
| RAVE | 2 | | | | | | | | |
| ControlVideo | 1 | 21.7GB | 6m28s | 23.5GB | 9m11s | — | — | — | — |
| Control-A-Video | 1 | 15.8GB | 42s | 16.6GB | 59s | 17.3GB | 2m22s | 18.5GB | 9m13s |
| TokenFlow | 1 | 10.9GB | 1m52s | 12.9GB | 3m17s | 15.5GB | 9m56s | 21.1GB | 47m33s |
| AdaFlow | 1 | 12.6GB | 1m07s | 15GB | 1m41s | 18.8GB | 5m14s | 23.82GB | 18m29s |
| GLFM | 1 | 10.0GB | 1m43s | 3m14s | 11.7GB | 22.1GB | 8m43s | 23.42GB | 41m20s |
| UELVE-Att(zero-shot) | 2 | | | | | | | | |
| UELVE-Att(one-shot) | 2 | | | | | | | | |
| UELVE-Att(LLM) | 2 | | | | | | | | |
| UELVE-CNet(zero-shot) | 2 | | | | | | | | |

# Response to Reviewer 3

Reviewer 3: The manuscript proposes UELVE, an LLM-driven unified framework for long-video editing. The method combines LLM-based prompt comparison and model adaptations, grid diffusion with a grid sliding manipulation, and multi-GPU data parallel computing. The topic is relevant and the proposed grid manipulation is interesting. However, several central claims are not sufficiently supported in the current version of the manuscript, leading me to recommend major revision, although my assessment is borderline between major revision and reject.

**Comment 3.1**

1. The contribution of the LLM is not sufficiently validated. The LLM is a central component of the proposed framework and is used to analyze source and target prompts and determine corresponding model adaptations, including local/global editing and zero-shot/one-shot settings. However, the ablation in Fig. 9 is limited to three qualitative examples and compares the LLM-based adaptive setting with two fixed configurations. A broader quantitative evaluation is needed to demonstrate that the LLM-based prompt analysis provides benefits across the considered editing tasks.

**Response 3.1:**

Table 6: Quantitative ablation study of the LLM routing module.

| Editing Task | Setting 1 (w/o LLM) | Setting 2 (w/o LLM) | Setting 3 (w/ LLM) |
|---|---|---|---|
| Local attribute editing | × | ✓ | ✓ |
| Object replacement | × | ✓ | ✓ |
| Global stylization | ✓ | × | ✓ |
| Motion editing | × | × | ✓ |
| Motion Preservation | × | ✓ | ✓ |

Thank you for your valuable comments. We appreciate your feedback. Admittedly, the three qualitative ablation cases in Figure9 alone cannot sufficiently verify the editing-quality benefits from LLM-based prompt analysis. We have therefore added quantitative ablation experiments for the LLM routing module, as reported in Table1.

Setting1 (w/o LLM) disables masking and temporal fine-tuning. While it performs well for global editing, it fails local editing and yields low Masked PSNR, SSIM and Tem-Con scores. Setting2 (w/o LLM) enables masking and temporal fine-tuning and suits local editing, yet suffers interference from redundant masks and fine-tuning for global editing. Despite metrics comparable to Setting3 (w/ LLM), it introduces extra overhead for mask generation and fine-tuning.

It is worth noting that improved editing quality is a side effect of the LLM module. Its primary role is to automatically adjust model configurations by analyzing prompts, so as to construct a unified video-editing framework that adapts to diverse editing tasks. The three examples in Figure 9 aim to highlight that the LLM can accomplish all three distinct types of editing tasks via flexible model-configuration adjustments, whereas fixed model configurations are only capable of partial editing tasks, as illustrated in Table 1.

**Comment 3.2**

Most of the ablation studies are only qualitative and are not convincing. In particular, the analyses of grid sliding (Fig. 7), grid size (Fig. 8), and the LLM component (Fig. 9) rely on a few visual examples, without quantitative measurements. Moreover, the differences shown in these figures are not always easy to assess due to their limited size and visual clarity. Since these components represent key elements of the proposed method, the authors should provide quantitative ablations over a sufficiently large set of videos, together with clearer visual comparisons.

**Response 3.2:**

**Comment 3.3**

Long-term temporal consistency is not directly demonstrated. The authors introduce the grid sliding manipulation to maintain long-term consistency. However, the metric "Tem-Con" is computed from CLIP visual embeddings of consecutive frames and therefore mainly reflects local temporal continuity. Stronger evidence is needed to show that the proposed method preserves consistency across distant portions of long videos.

**Response 3.3:**

**Comment 3.4**

The preliminary section is unnecessarily lengthy. Section 3.2 devotes substantial space to standard background on latent diffusion, DDPMs, and DDIMs, including forward/backward diffusion formulations and the training objective. These concepts are well-established for the expected readership of the journal. I suggest shortening this section and retaining only the notation and equations directly needed to describe proposed method. The saved space could be better used to clarify the methodological and implementation details.

**Response 3.4:**

**Comment 3.5**

The experimental validation of the unified framework is limited. A main claim of the paper is that UELVE can adapt to different editing tasks through LLM-based prompt analysis and corresponding model adaptations. However, this capability is mainly illustrated qualitatively in Fig. 9. The quantitative results in Table 1 are reported for UELVE-Attention and UELVE-ControlNet, but do not clearly show how the LLM-based adaptations contribute across the different editing categories. A task-wise quantitative analysis would better support the claim that the same framework can effectively handle diverse editing scenarios.

**Response 3.5:**

Table 7: Per-task quantitative ablation study for LLM-driven adaptation. All settings share the identical grid-diffusion backbone. Setting 1: fixed zero-shot without mask and fine-tuning; Setting 2: fixed one-shot with mask and temporal module fine-tuning always enabled; Setting 3: full UELVE with LLM-guided adaptive configuration selection; Setting 4: fixed one-shot with mask and Motion module fine-tuning always enabled.

| Editing Task | Settings | Masked-PSNR ↑ | SSIM ↑ | Clip-T ↑ | Tem-Con ↑ |
|---|---|---|---|---|---|
| Global stylization | Setting1 | | | | |
| | Setting2 | | | | |
| | Setting3 | | 0.668 | 0.286 | 0.970 |
| | Setting4 | — | — | — | — |
| Local Attribute Modification | Setting1 | | | | |
| | Setting2 | 284 | 071 | 0.287 | 0.966 |
| | Setting3 | **24.01** | **0.676** | **0.284** | **0.968** |
| | Setting4 | — | — | — | — |
| Complex Structure Change | Setting1 | | | | |
| | Setting2 | | | | |
| | Setting3 | | | | |
| | Setting4 | — | — | — | — |
| Motion Preservation | Setting1 | | | | |
| | Setting2 | | | | |
| | Setting3 | | | | |
| | Setting4 | — | — | — | — |
| Motion Editing | Setting1 | — | — | — | — |
| | Setting2 | — | — | — | — |
| | Setting3 | | | | |
| | Setting4 | | 0.668 | 0.286 | 0.970 |

**Comment 3.6**

The authors do not provide code or an implementation repository. Given the high complexity of the approach, this limits reproducibility. The authors should release the code and provide more implementation details sufficient to reproduce the reported results.

**Response 3.6:**

**Comment 3.7**

The long-video editing claim requires stronger experimental support. The datasets described in the paper include videos lasting approximately one minute and self-captured videos of up to several minutes, whereas the main quantitative evaluation considers at most 200 frames. Given that long-video editing is a central motivation and contribution of the work, experiments on substantially longer sequences would be necessary to support this claim.

**Response 3.7:**

**Comment 3.8**

8. The mathematical notation in Section 3.2 should be carefully revised. The manuscript also contains numerous grammatical and typographical errors. Several figures contain very small text, and labels such as "GUP1/GUP2" in Fig. 5 should be corrected.

**Response 3.8:**

Thank you for pointing out the writing and presentation issues. In response to your suggestions, we have thoroughly proofread the entire manuscript. We have corrected all typos, grammatical and punctuation errors, as well as non-standard usages. Moreover, we have standardized the formatting of technical terms and polished the academic writing to improve the overall readability of the paper.

# Response to Reviewer 4

Reviewer 3: The paper addresses text guided long video editing, an active and important topic in the diffusion and video generation field. The grid sliding mechanism and multi GPU scheme are likely reusable by other researchers, while the LLM routing component is more specific to this pipeline and may already be less competitive given the growing adoption of vision language models for direct visual guidance.

**Comment 4.1**

Technically sound overall, but lacks statistical reporting (variance across runs, confidence intervals, significance tests) in Tables 1 to 3.

**Response 4.1:**

The variance across runs, confidence intervals and significance tests for Table 1.

Table 8: The variance across runs for Table1, Mean ± standard deviation for every cell.

| Method | Masked PSNR↑ | | | | SSIM↑ | | | | Clip-T↑ | | | | Tem-Con↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Tune-A-Video | 13.91 ± 0.12 | 15.87 ± 0.14 | 15.13 ± 0.12 | — | 0.4350 ± 0.0239 | 0.4749 ± 0.0170 | 0.4418 ± 0.0110 | — | 0.2960 ± 0.0048 | 0.2727 ± 0.0080 | 0.2894 ± 0.0077 | — | 0.9488 ± 0.0039 | 0.9481 ± 0.0075 | 0.9441 ± 0.0109 | — |
| Video-P2P | 23.66 ± 0.23 | 22.23 ± 0.21 | — | — | 0.6563 ± 0.0226 | 0.6217 ± 0.0202 | — | — | 0.2937 ± 0.0123 | 0.2955 ± 0.0079 | — | — | 0.9695 ± 0.0061 | 0.9640 ± 0.0109 | — | — |
| FateZero | 23.83 ± 0.19 | — | — | — | 0.6572 ± 0.0166 | — | — | — | 0.2872 ± 0.0119 | — | — | — | 0.9576 ± 0.0124 | — | — | — |
| RAVE | 22.98 ± 0.16 | 20.18 ± 0.20 | 20.24 ± 0.12 | 20.16 ± 0.16 | 0.5680 ± 0.0230 | 0.5349 ± 0.0073 | 0.5894 ± 0.0103 | 0.5419 ± 0.0120 | 0.3145 ± 0.0085 | 0.3128 ± 0.0082 | 0.3044 ± 0.0127 | 0.2928 ± 0.0141 | 0.9795 ± 0.0171 | 0.9753 ± 0.0074 | 0.9696 ± 0.0147 | 0.9619 ± 0.0088 |
| ControlVideo | 20.60 ± 0.15 | 19.97 ± 0.17 | — | — | 0.5923 ± 0.0210 | 0.5517 ± 0.0180 | — | — | 0.3070 ± 0.0074 | 0.3012 ± 0.0091 | — | — | 0.9746 ± 0.0082 | 0.9732 ± 0.0067 | — | — |
| Control-A-Video | 10.72 ± 0.14 | 9.715 ± 0.18 | 11.52 ± 0.13 | — | 0.4802 ± 0.0221 | 0.3944 ± 0.0242 | 0.4245 ± 0.0131 | — | 0.2942 ± 0.0062 | 0.3065 ± 0.0078 | 0.2979 ± 0.0083 | — | 0.9673 ± 0.0071 | 0.9638 ± 0.0093 | 0.9506 ± 0.0112 | — |
| TokenFlow | 23.55 ± 0.08 | 20.71 ± 0.27 | 20.03 ± 0.29 | 20.38 ± 0.27 | 0.6502 ± 0.0198 | 0.6431 ± 0.0127 | 0.6313 ± 0.0096 | 0.6130 ± 0.0149 | 0.2917 ± 0.0098 | 0.2980 ± 0.0090 | 0.2973 ± 0.0072 | 0.2898 ± 0.0099 | 0.9776 ± 0.0160 | 0.9712 ± 0.0104 | 0.9724 ± 0.0132 | 0.9718 ± 0.0055 |
| AdaFlow | 23.10 ± 0.19 | 21.15 ± 0.27 | 20.15 ± 0.18 | 19.32 ± 0.08 | 0.6500 ± 0.0261 | 0.6499 ± 0.0201 | 0.6357 ± 0.0129 | 0.5819 ± 0.0209 | 0.2945 ± 0.0065 | 0.2961 ± 0.0071 | 0.3040 ± 0.0072 | 0.2708 ± 0.0121 | 0.9789 ± 0.0119 | 0.9762 ± 0.0070 | 0.9720 ± 0.0109 | 0.9517 ± 0.0094 |
| GLFM | 22.91 ± 0.16 | 20.49 ± 0.18 | 20.05 ± 0.23 | 20.19 ± 0.19 | 0.6031 ± 0.0064 | 0.6190 ± 0.0149 | 0.6217 ± 0.0204 | 0.6017 ± 0.0133 | 0.3149 ± 0.0152 | 0.3117 ± 0.0070 | 0.3037 ± 0.0091 | 0.2991 ± 0.0073 | 0.9791 ± 0.0139 | 0.9677 ± 0.0104 | 0.9626 ± 0.0163 | 0.9563 ± 0.0055 |
| UELVE-Attention | **24.05 ± 0.13** | **22.83 ± 0.18** | **21.78 ± 0.09** | **21.65 ± 0.18** | **0.6849 ± 0.0210** | **0.6635 ± 0.0200** | 0.6176 ± 0.0187 | 0.5833 ± 0.0074 | 0.2841 ± 0.0047 | 0.2955 ± 0.0051 | 0.2922 ± 0.0056 | 0.2893 ± 0.0076 | 0.9653 ± 0.0058 | 0.9655 ± 0.0123 | 0.9594 ± 0.0124 | **0.9725 ± 0.0139** |
| UELVE-ControlNet | 21.18 ± 0.15 | 20.00 ± 0.27 | 20.29 ± 0.25 | 20.47 ± 0.09 | 0.5711 ± 0.0290 | 0.5853 ± 0.0274 | 0.5412 ± 0.0185 | 0.5570 ± 0.0135 | **0.3174 ± 0.0145** | **0.3210 ± 0.0095** | **0.3056 ± 0.0088** | 0.2957 ± 0.0050 | **0.9801 ± 0.0028** | **0.9765 ± 0.0095** | 0.9707 ± 0.0083 | 0.9643 ± 0.0106 |

Table 9: 95% t-distribution confidence interval of each method.

| Method | Masked PSNR↑ | | | | SSIM↑ | | | | Clip-T↑ | | | | Tem-Con↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Tune-A-Video | [13.76, 14.06] | [15.69, 16.05] | [14.98, 15.28] | — | [0.405, 0.465] | [0.454, 0.496] | [0.428, 0.456] | — | [0.290, 0.302] | [0.263, 0.282] | [0.280, 0.299] | — | [0.943, 0.955] | [0.939, 0.957] | [0.931, 0.957] | — |
| Video-P2P | [23.37, 23.95] | [21.96, 22.50] | — | — | [0.628, 0.684] | [0.590, 0.653] | — | — | [0.278, 0.309] | [0.286, 0.305] | — | — | [0.962, 0.977] | [0.951, 0.977] | — | — |
| FateZero | [23.59, 24.07] | — | — | — | [0.637, 0.677] | — | — | — | [0.273, 0.301] | — | — | — | [0.942, 0.973] | — | — | — |
| RAVE | [22.78, 23.18] | [19.93, 20.43] | [20.09, 20.39] | [19.96, 20.36] | [0.536, 0.600] | [0.525, 0.545] | [0.575, 0.604] | [0.527, 0.557] | [0.304, 0.325] | [0.302, 0.323] | [0.289, 0.320] | [0.275, 0.311] | [0.958, 1.000] | [0.965, 0.986] | [0.949, 0.990] | [0.950, 0.974] |
| ControlVideo | [20.41, 20.79] | [19.75, 20.19] | — | — | [0.563, 0.622] | [0.527, 0.576] | — | — | [0.298, 0.316] | [0.290, 0.312] | — | — | [0.964, 0.985] | [0.963, 0.983] | — | — |
| Control-A-Video | [10.54, 10.90] | [9.48, 9.95] | [11.35, 11.69] | — | [0.449, 0.511] | [0.360, 0.429] | [0.405, 0.444] | — | [0.287, 0.301] | [0.296, 0.317] | [0.288, 0.308] | — | [0.957, 0.978] | [0.951, 0.977] | [0.936, 0.965] | — |
| TokenFlow | [23.45, 23.65] | [20.37, 21.05] | [19.67, 20.39] | [20.04, 20.72] | [0.626, 0.674] | [0.628, 0.658] | [0.619, 0.644] | [0.593, 0.633] | [0.279, 0.304] | [0.286, 0.310] | [0.288, 0.307] | [0.278, 0.302] | [0.957, 0.998] | [0.958, 0.984] | [0.956, 0.989] | [0.964, 0.980] |
| AdaFlow | [22.87, 23.33] | [20.81, 21.49] | [19.92, 20.38] | [19.22, 19.42] | [0.618, 0.682] | [0.625, 0.675] | [0.619, 0.653] | [0.557, 0.607] | [0.287, 0.302] | [0.287, 0.305] | [0.296, 0.312] | [0.259, 0.283] | [0.965, 0.993] | [0.967, 0.985] | [0.960, 0.984] | [0.941, 0.962] |
| GLFM | [22.72, 23.10] | [20.27, 20.71] | [19.78, 20.32] | [19.95, 20.43] | [0.594, 0.612] | [0.601, 0.637] | [0.589, 0.654] | [0.583, 0.621] | [0.294, 0.336] | [0.303, 0.321] | [0.293, 0.314] | [0.289, 0.309] | [0.960, 0.998] | [0.954, 0.981] | [0.943, 0.982] | [0.948, 0.964] |
| UELVE-Attention | [**23.89**, **24.21**] | [**22.55**, **23.11**] | [**21.67**, **21.89**] | [**21.38**, **21.92**] | [**0.659**, **0.711**] | [**0.637**, **0.690**] | [0.594, 0.641] | [0.573, 0.593] | [0.278, 0.290] | [0.289, 0.302] | [0.285, 0.299] | [0.281, 0.298] | [0.958, 0.973] | [0.950, 0.981] | [0.944, 0.975] | [**0.955**, **0.990**] |
| UELVE-ControlNet | [21.00, 21.36] | [19.58, 20.42] | [19.84, 20.74] | [20.33, 20.61] | [0.533, 0.609] | [0.545, 0.626] | [0.516, 0.566] | [0.539, 0.575] | [**0.297**, **0.338**] | [**0.307**, **0.335**] | [**0.297**, **0.314**] | [0.288, 0.303] | [**0.976**, **0.984**] | [**0.963**, **0.990**] | [0.960, 0.981] | [0.951, 0.977] |

Since our UELVE has two implementations, we present the results of significance tests separately.

The variance across runs and confidence intervals for Table 2. Runtime metrics shown in Table 2 and Table 3 represent wall-clock inference time. Significance tests (p-values) are not required for runtime measurements.

The variance across runs and confidence intervals for Table 3.

**Comment 4.2**

Technical depth is uneven: the grid diffusion and multi GPU sections are formally detailed, while the ablation studies remain mostly qualitative.

**Response 4.2:**

Thank you for your positive feedback on our work. We also recognize that our ablation section lacks support from quantitative experimental results. As shown in Table 1 above, we present quantitative ablation experiments for grid-sliding and grid-size.

Table 10: Pairwise p-values from paired two-sided t-test. Each cell reports p-value comparing UELVE-Attention against the baseline on the corresponding row. Note that p-value is a pairwise-comparison statistic rather than intrinsic property of a single method. $p < 0.05$ indicates statistically significant difference.

| **Method** | **Masked PSNR↑** | | | | **SSIM↑** | | | | **Clip-T↑** | | | | **Tem-Con↑** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Tune-A-Video | < 0.001 | < 0.001 | < 0.001 | — | < 0.001 | < 0.001 | < 0.001 | — | 0.002 | 0.002 | 0.074 | — | 0.005 | 0.185 | 0.002 | — |
| Video-P2P | 0.115 | 0.028 | — | — | 0.764 | 0.127 | — | — | 0.029 | 0.031 | — | — | 0.072 | 0.800 | — | — |
| FateZero | 0.525 | — | — | — | 0.452 | — | — | — | 0.474 | — | — | — | 0.652 | — | — | — |
| RAVE | < 0.001 | < 0.001 | < 0.001 | < 0.001 | 0.002 | < 0.001 | 0.042 | 0.044 | 0.003 | 0.093 | 0.009 | 0.523 | 0.142 | 0.016 | 0.801 | 0.608 |
| ControlVideo | < 0.001 | < 0.001 | — | — | 0.003 | 0.008 | — | — | 0.011 | 0.022 | — | — | 0.033 | 0.041 | — | — |
| Control-A-Video | < 0.001 | < 0.001 | < 0.001 | — | < 0.001 | < 0.001 | < 0.001 | — | 0.004 | 0.007 | 0.013 | — | 0.008 | 0.014 | 0.021 | — |
| TokenFlow | 0.006 | < 0.001 | < 0.001 | < 0.001 | 0.191 | 0.154 | 0.716 | 0.004 | 0.086 | 0.132 | 0.002 | 0.867 | 0.070 | 0.384 | 0.232 | 0.139 |
| AdaFlow | 0.005 | < 0.001 | < 0.001 | < 0.001 | 0.111 | 0.457 | 0.783 | 0.317 | < 0.001 | 0.003 | 0.009 | 0.040 | 0.063 | 0.392 | 0.402 | 0.404 |
| GLFM | < 0.001 | < 0.001 | < 0.001 | < 0.001 | 0.001 | 0.047 | 0.687 | 0.025 | 0.022 | 0.006 | 0.002 | 0.012 | 0.076 | 0.642 | 0.893 | 0.119 |
| UELVE-Attention | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |
| UELVE-ControlNet | < 0.001 | < 0.001 | < 0.001 | < 0.001 | 0.005 | 0.011 | 0.002 | 0.583 | 0.012 | 0.009 | 0.003 | 0.060 | 0.002 | 0.491 | 0.872 | 0.535 |

Table 11: Pairwise p-values from paired two-sided t-test. Each cell reports p-value comparing UELVE-ControlNet against the baseline on the corresponding row.

| **Method** | **Masked PSNR↑** | | | | **SSIM↑** | | | | **Clip-T↑** | | | | **Tem-Con↑** | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Tune-A-Video | < 0.001 | < 0.001 | < 0.001 | — | < 0.001 | 0.002 | < 0.001 | — | 0.101 | < 0.001 | 0.054 | — | < 0.001 | 0.001 | 0.026 | — |
| Video-P2P | < 0.001 | < 0.001 | — | — | 0.006 | 0.097 | — | — | 0.156 | 0.005 | — | — | 0.286 | 0.314 | — | — |
| FateZero | < 0.001 | — | — | — | 0.003 | — | — | — | 0.015 | — | — | — | 0.035 | — | — | — |
| RAVE | < 0.001 | 0.384 | 0.668 | 0.010 | 0.210 | 0.007 | 0.008 | 0.046 | 0.545 | 0.095 | 0.874 | 0.058 | 0.776 | 0.063 | 0.695 | 0.761 |
| ControlVideo | 0.042 | 0.812 | — | — | 0.088 | 0.124 | — | — | 0.331 | 0.066 | — | — | 0.442 | 0.537 | — | — |
| Control-A-Video | < 0.001 | < 0.001 | < 0.001 | — | < 0.001 | < 0.001 | < 0.001 | — | 0.032 | < 0.001 | 0.022 | — | 0.001 | 0.003 | 0.007 | — |
| TokenFlow | < 0.001 | 0.027 | 0.037 | 0.173 | 0.001 | 0.009 | 0.001 | 0.001 | 0.097 | < 0.001 | 0.296 | 0.348 | 0.857 | 0.920 | 0.123 | 0.058 |
| AdaFlow | < 0.001 | 0.001 | 0.532 | < 0.001 | 0.011 | 0.007 | 0.004 | 0.226 | 0.130 | 0.001 | 0.790 | 0.012 | 0.850 | 0.899 | 0.392 | 0.428 |
| GLFM | < 0.001 | 0.056 | 0.086 | 0.023 | 0.064 | 0.128 | 0.013 | 0.027 | 0.950 | 0.352 | 0.260 | 0.044 | 0.290 | 0.860 | 0.958 | 0.199 |
| UELVE-Attention | < 0.001 | < 0.001 | < 0.001 | < 0.001 | 0.005 | 0.011 | 0.002 | 0.583 | 0.012 | 0.009 | 0.003 | 0.060 | 0.002 | 0.491 | 0.872 | 0.535 |
| UELVE-ControlNet | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — | — |

**Comment 4.3**

Reference to earlier work is insufficient in one respect: "Toward General-Purpose Video Reconstruction through Synergy of Grid-Splicing Diffusion and Large Language Models" (IEEE TCSVT February 2026) applies the same grid diffusion principle with the same motivation and analogy, combined with a vision language model for guidance, but is not cited or discussed. The authors should add this reference and clarify their contribution relative to it, particularly the choice of a text only LLM over a vision language model.

**Response 4.3:**

Thank you for pointing out this relevant TCSVT reference. We have added the citation "Toward General-Purpose Video Reconstruction through Synergy of Grid-Splicing Diffusion and Large Language Models" (IEEE TCSVT, Feb 2026) into our manuscript, and added a dedicated discussion in Related Work to clarify the differences between our work and this paper.

Although both works adopt grid based diffusion schemes to process long videos, there exist fundamental differences in task objective, grid mechanism design, and the role of large models: 1) Task difference: The TCSVT work targets video reconstruction / enhancement (restoration task). In contrast, our UELVE focuses on text-guided long video editing. 2) Grid mechanism: The grid-splicing scheme in the TCSVT work is designed primarily to reduce

Table 12: Mean ± standard deviation corresponding to main paper Table 2.

| **Method** | 8-f | 16-f | 48-f | 200-f |
|---|---|---|---|---|
| | (sec) | (sec) | (sec) | (sec) |
| Tune-A-Video | $19.0 \pm 0.7$ | $35.0 \pm 1.1$ | $102.0 \pm 2.4$ | — |
| Video-P2P | $61.0 \pm 1.6$ | $124.0 \pm 3.0$ | — | — |
| FateZero | $145.0 \pm 3.3$ | — | — | — |
| RAVE ×2GPU | $19.0 \pm 0.6$ | $37.0 \pm 1.0$ | $104.0 \pm 2.2$ | $429.0 \pm 7.1$ |
| ControlVideo | $62.0 \pm 1.7$ | $250.0 \pm 4.4$ | — | — |
| Control-A-Video | $25.0 \pm 0.8$ | $46.0 \pm 1.2$ | $123.0 \pm 2.6$ | $492.0 \pm 7.8$ |
| TokenFlow | $51.0 \pm 1.3$ | $108.0 \pm 2.3$ | $334.0 \pm 5.5$ | $1301.0 \pm 14.2$ |
| AdaFlow | $34.0 \pm 0.9$ | $72.0 \pm 1.6$ | $215.0 \pm 3.8$ | $901.0 \pm 11.3$ |
| GLFM | $51.0 \pm 1.2$ | $115.0 \pm 2.4$ | $350.0 \pm 5.8$ | $1266.0 \pm 13.8$ |
| UELVE-Attention ×2GPU | $33.0 \pm 0.8$ | $63.0 \pm 1.4$ | $189.0 \pm 3.2$ | $825.0 \pm 10.4$ |
| UELVE-ControlNet ×2GPU | $18.0 \pm 0.5$ | $34.0 \pm 0.9$ | $94.0 \pm 2.0$ | $372.0 \pm 6.3$ |

GPU memory consumption for long video reconstruction. It packs patches from multiple frames into a static grid and relies on self-attention within this concatenated grid image for implicit cross-patch information flow, without explicit grid-sliding design to enforce long-range consistency across separated grids. In our UELVE, besides memory-friendly grid partitioning, we further propose grid sliding manipulation, which alternates grid layouts across diffusion timesteps to explicitly model cross-grid interactions for long-term temporal coherence. 3) Rationale for text only LLM instead of vision-language-model (VLM): The TCSVT paper employs a VLM, since its reconstruction task requires visual perception over input video frames to identify visual artifacts and guide restoration. However, our text-driven video editing task derives all editing intent purely from the semantic difference between source text prompt and target text prompt. In our pipeline, the LLM acts as a decision making agent. it only performs text-level comparison and outputs editing strategy decisions. Our LLM never observes video frames, so visual understanding capability from VLM is unnecessary for our core workflow. Adopting VLM would introduce extra visual encoding overhead and potential visual noise that may perturb strategy decisions derived from prompt semantics. Therefore we choose a text-only LLM for our framework. We also note that integrating VLM for more complex multi-modal editing can be a promising direction for future extensions, which is out-of-scope for the current paper.

Table 13: 95% t-distribution confidence interval corresponding to main paper Table 2.

| **Method** | 8-f | 16-f | 48-f | 200-f |
|---|---|---|---|---|
| | (sec) | (sec) | (sec) | (sec) |
| Tune-A-Video | [18.1, 19.9] | [33.6, 36.4] | [99.2, 104.8] | — |
| Video-P2P | [59.0, 63.0] | [120.3, 127.7] | — | — |
| FateZero | [140.3, 149.7] | — | — | — |
| RAVE ×2GPU | [18.2, 19.8] | [35.7, 38.3] | [101.4, 106.6] | [420.6, 437.4] |
| ControlVideo | [59.9, 64.1] | [244.6, 255.4] | — | — |
| Control-A-Video | [24.1, 25.9] | [44.5, 47.5] | [120.0, 126.0] | [482.8, 501.2] |
| TokenFlow | [49.3, 52.7] | [105.0, 111.0] | [328.0, 340.0] | [1284.4, 1317.6] |
| AdaFlow | [32.8, 35.2] | [69.8, 74.2] | [210.8, 219.2] | [887.8, 914.2] |
| GLFM | [49.4, 52.6] | [111.9, 118.1] | [343.7, 356.3] | [1249.9, 1282.1] |
| UELVE-Attention ×2GPU | [31.9, 34.1] | [60.8, 65.2] | [185.3, 192.7] | [812.8, 837.2] |
| UELVE-ControlNet ×2GPU | [17.3, 18.7] | [32.7, 35.3] | [91.5, 96.5] | [364.6, 379.4] |

Table 14: Mean ± standard deviation corresponding to main paper Table 3.

| **Method** | Inv-8f | Den-8f | Inv-16f | Den-16f | Inv-48f | Den-48f | Inv-200f | Den-200f |
|---|---|---|---|---|---|---|---|---|
| Video-Infinity | — | 12.0 ± 0.4 | — | 18.0 ± 0.5 | — | 48.0 ± 1.1 | — | — |
| Video-Infinity ×2GPU | — | 22.0 ± 0.6 | — | 36.0 ± 0.9 | — | 40.0 ± 1.0 | — | 103.0 ± 2.1 |
| RAVE | 17.0 ± 0.5 | 20.0 ± 0.6 | 30.0 ± 0.8 | 39.0 ± 0.9 | 83.0 ± 1.8 | 107.0 ± 2.2 | 322.0 ± 5.3 | 448.0 ± 6.6 |
| RAVE ×2GPU | 9.0 ± 0.3 | 10.0 ± 0.4 | 17.0 ± 0.5 | 20.0 ± 0.6 | 47.0 ± 1.2 | 57.0 ± 1.4 | 188.0 ± 3.7 | 241.0 ± 4.5 |
| UELVE-Attention | 24.0 ± 0.7 | 34.0 ± 0.9 | 57.0 ± 1.3 | 61.0 ± 1.4 | 164.0 ± 3.1 | 196.0 ± 3.6 | 688.0 ± 9.2 | 865.0 ± 10.7 |
| UELVE-Attention ×2GPU | 15.0 ± 0.5 | 18.0 ± 0.6 | 29.0 ± 0.7 | 34.0 ± 0.8 | 88.0 ± 1.8 | 101.0 ± 2.1 | 366.0 ± 5.8 | 459.0 ± 6.9 |
| UELVE-ControlNet | 16.0 ± 0.5 | 18.0 ± 0.6 | 29.0 ± 0.7 | 36.0 ± 0.8 | 72.0 ± 1.6 | 98.0 ± 2.0 | 316.0 ± 5.1 | 367.0 ± 5.7 |
| UELVE-ControlNet ×2GPU | 9.0 ± 0.3 | 9.0 ± 0.3 | 16.0 ± 0.4 | 18.0 ± 0.5 | 40.0 ± 1.1 | 54.0 ± 1.3 | 170.0 ± 3.4 | 202.0 ± 4.0 |

**Comment 4.4**

The title adequately reflects the content and keywords are reasonably chosen. The abstract is purely qualitative and should include key quantitative results. The introduction explains motivation and contributions, but could be more compact. The paper is well organised and of appropriate length overall, though the ablation section is thin relative to the quantitative rigor shown elsewhere. English is satisfactory with minor lapses. Presentation quality is above average, with clear figures and tables and no unnecessary duplication, though the color choice in Figure 2 (teal and orange text) reduces readability and should be revised.

**Response 4.4:**

Table 15: 95% t-distribution confidence interval corresponding to main paper Table 3.

| **Method** | Inv-8f | Den-8f | Inv-16f | Den-16f | Inv-48f | Den-48f | Inv-200f | Den-200f |
|---|---|---|---|---|---|---|---|---|
| Video-Infinity | — | [11.5, 12.5] | — | [17.4, 18.6] | — | [46.6, 49.4] | — | — |
| Video-Infinity ×2GPU | — | [21.2, 22.8] | — | [34.8, 37.2] | — | [38.8, 41.2] | — | [100.4, 105.6] |
| RAVE | [16.4, 17.6] | [19.3, 20.7] | [29.1, 30.9] | [38.0, 40.0] | [80.8, 85.2] | [104.3, 109.7] | [315.4, 328.6] | [439.8, 456.2] |
| RAVE ×2GPU | [8.6, 9.4] | [9.5, 10.5] | [16.4, 17.6] | [19.3, 20.7] | [45.5, 48.5] | [55.3, 58.7] | [183.4, 192.6] | [235.4, 246.6] |
| UELVE-Attention | [23.1, 24.9] | [32.9, 35.1] | [55.4, 58.6] | [59.3, 62.7] | [160.1, 167.9] | [191.4, 200.6] | [676.5, 699.5] | [851.6, 878.4] |
| UELVE-Attention ×2GPU | [14.4, 15.6] | [17.3, 18.7] | [28.1, 29.9] | [33.0, 35.0] | [85.5, 90.5] | [98.4, 103.6] | [358.7, 373.3] | [450.3, 467.7] |
| UELVE-ControlNet | [15.4, 16.6] | [17.3, 18.7] | [28.2, 29.8] | [35.1, 36.9] | [70.2, 73.8] | [95.7, 100.3] | [309.7, 322.3] | [360.0, 374.0] |
| UELVE-ControlNet ×2GPU | [8.6, 9.4] | [8.6, 9.4] | [15.5, 16.5] | [17.4, 18.6] | [38.7, 41.3] | [52.4, 55.6] | [165.6, 174.4] | [196.8, 207.2] |

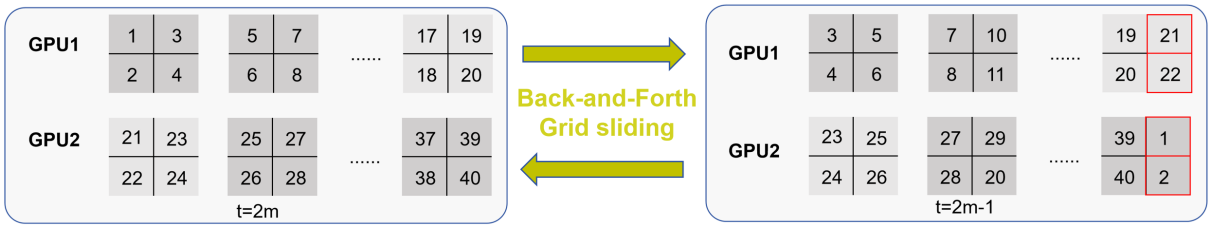



Figure 1: layout2

# Response to Reviewer 5

Reviewer 3: The manuscript proposes UELVE, an LLM-driven unified framework for efficient long-video editing using grid diffusion, grid sliding, and multi-GPU parallelism. The topic is relevant and the proposed framework shows promising results. However, several important issues should be further addressed before publication.

**Comment 5.1**

The contribution of the LLM module is insufficiently validated. The LLM mainly acts as a router to determine local/global editing and zero-/one-shot configurations according to predefined guidelines. The current evaluation is largely qualitative. Please provide quantitative evaluation of the routing accuracy and comparisons with rule-based, manual, or oracle strategy selection. Robustness to different prompts and LLMs should also be investigated.

**Response 5.1:**

**Comment 5.2**

The long-video claim requires stronger experimental support. Although the datasets contain minute-level videos, the main quantitative evaluations are limited to at most 200 frames. Please evaluate the method on substantially longer, preferably full minute-level videos, and report editing quality, temporal consistency, memory consumption, runtime, and performance degradation as video length increases.

**Response 5.2:**

**Comment 5.3**

The ablations of grid sliding and grid size are mainly qualitative. Since grid sliding is a central technical contribution, quantitative comparisons of grid diffusion with/without sliding, different grid sizes, and different sliding configurations should be provided using temporal consistency and editing-quality metrics.

**Response 5.3:**

Thank you for your comments. In addition to the grid sliding configuration (Layout 1) presented in the main text, we designed two other grid sliding strategies, as illustrated in 1

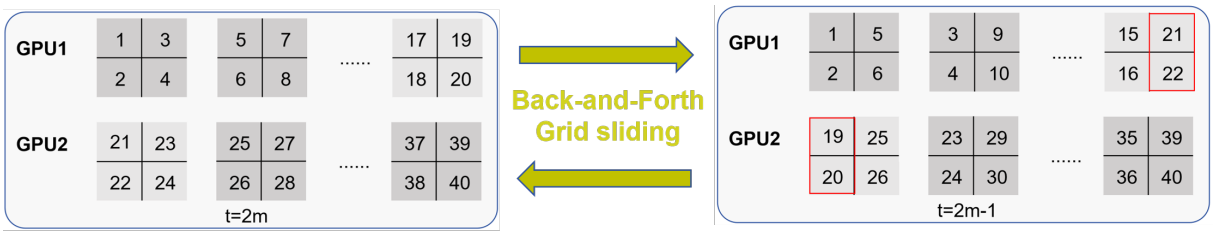



Figure 2: layout3

Table 16: The ablation study of gird sliding mechanism.

| Method | Masked PSNR↑ | | | | SSIM↑ | | | | Clip-T↑ | | | | Tem-Con↑ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f | 8-f | 16-f | 48-f | 200-f |
| Layout1(Attention) [**TAV**] | 13.91 | 15.87 | 15.13 | — | 0.4350 | 0.4749 | 0.4418 | — | 0.2960 | 0.2727 | 0.2894 | — | 0.9488 | 0.9481 | 0.9441 | — |
| Layout1(ControlNet) [**TAV**] | 13.91 | 15.87 | 15.13 | — | 0.4350 | 0.4749 | 0.4418 | — | 0.2960 | 0.2727 | 0.2894 | — | 0.9488 | 0.9481 | 0.9441 | — |
| Layout2(Attention) | 23.66 | 22.23 | — | — | 0.6563 | 0.6217 | — | — | 0.2937 | 0.2955 | — | — | 0.9695 | 0.9640 | — | — |
| Layout2(ControlNet) | 23.66 | 22.23 | — | — | 0.6563 | 0.6217 | — | — | 0.2937 | 0.2955 | — | — | 0.9695 | 0.9640 | — | — |
| Layout3(Attention) | 23.83 | — | — | — | 0.6572 | — | — | — | 0.2872 | — | — | — | 0.9576 | — | — | — |
| Layout3(ControlNet) | 23.83 | — | — | — | 0.6572 | — | — | — | 0.2872 | — | — | — | 0.9576 | — | — | — |

and 2.

Layout 2 connects the start and end segments during editing. It not only eliminates empty grid regions at the two ends that occur in Layout 1 during grid sliding, but also further enhances consistency between the beginning and end to mitigate consistency drift. For short videos, the content at the start and end generally does not differ much from the main body, making Layout 2 suitable for such cases. In contrast, long videos usually exhibit substantial differences between their opening and closing segments, where such start-end connections are unnecessary. Therefore, Layout 2 is tailored for short-video editing.

Built upon Layout 1, Layout 3 strengthens cross-grid interactions. This grants Layout 3 superior capability in preserving long-range consistency, rendering it more suitable for long videos. Nevertheless, when applied to short videos, the excessive interactions in Layout 3 introduce undesirable mutual interference among content regions and degrade editing quality. By comparison, Layout 1 achieves a balanced performance across both short and long video editing tasks. For this reason, we adopt Layout 1 as the default configuration.

**Comment 5.4**

The LLM only observes the source and target text prompts. Some decisions, particularly motion-related editing complexity, may depend strongly on the visual and temporal content of the source video. Please discuss this limitation and evaluate cases in which similar textual prompts correspond to videos with substantially different motion complexity.

**Response 5.4:**

Thank you for your insightful feedback. We fully acknowledge this critical limitation. Our LLM makes decisions by comparing source and target prompts. It works well when prompts faithfully describe the video, yet may yield inaccurate judgements if prompts are vague or cannot capture real-world video characteristics.

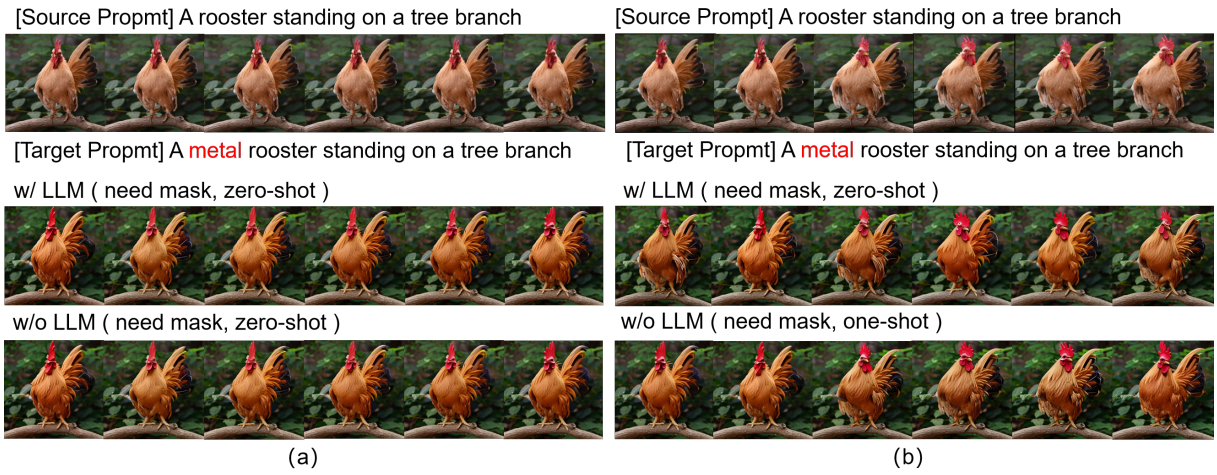



Figure 3: Enter Caption

As shown in Fig 3, Consider two source videos with identical prompt pairs, source prompt: A rooster standing on a tree branch, target prompt: A metal rooster standing on a tree branch. For the video 1 in (a) the rooster is nearly static; the video 2 in (b) the rooster continuously turns its head left and right. Despite substantially different motion complexity, the LLM outputs the same decision: need mask and zero-shot.

For static video 1 (a), this decision matches human preference and produces good results. For video 2 (b) with frequent head motion, human oracle would select need mask and one-shot. Without temporal fine-tuning under LLM's zero-shot setting, temporal consistency will degrade, especially on the rooster's head across frames.

In future work, we will introduce lightweight video motion statistics (e.g. optical-flow magnitude, frame-wise CLIP feature distance) as auxiliary input for the LLM agent. Fast-inference VLMs may also be adopted to support visual-grounded decision-making.

**Comment 5.5**

The implementation of the grid representation needs clarification. All frames are reported to be resized to 512×512, while multiple frames are further arranged into 2×2 or 3×3 grids. Please clarify the final grid resolution, effective per-frame resolution, resizing strategy, treatment of incomplete grids, and whether all baselines are evaluated at the same effective resolution.

**Response 5.5:**

Thank you for pointing out these missing implementation details about our grid representation. We acknowledge that these details were not sufficiently clarified in the original manuscript. Each raw video frame is pre-processed to 512×512 by first resizing the short side to 512 with bilinear interpolation and anti-aliasing, followed by center-cropping. When assembling frames into an $n \times l$ grid, each individual frame retains its original 512×512 resolution; the stitched grid image therefore has a pixel resolution of $(512n) \times (512l)$, corresponding to a VAE latent resolution of $(64n) \times (64l)$. For example, a 2×2 grid yields a 1024×1024 pixel image with 128×128 latent representation. Frame stitching is performed via lossless pixel block copying without additional resampling, so the effective resolution per frame remains

512×512. After latent denoising, we reverse the stitching operation to recover individual 512×512 output frames. For video sequences whose total frame count cannot be perfectly divided by the grid capacity $n \cdot l$, we pad the final incomplete grid by replicating the last valid frame of the video to form a full grid input; these padded duplicate frames are discarded after decoding and are not present in the final edited video. For fair comparison, all baseline methods take exactly the same pre-processed 512×512 frames as input and run under their official default 512×512 resolution setting.

**Comment 5.6**

The efficiency comparison should report complete end-to-end cost. The current runtime mainly includes inversion and denoising, while UELVE may additionally involve LLM inference, mask generation, and per-video fine-tuning. These costs should be included when supporting the claim of efficient and cost-effective unified editing.

**Response 5.6:**

**Comment 5.7**

Some experimental claims should be stated more carefully. Table 1 shows that UELVE does not achieve the best result for every metric and configuration. Task-specific results for local, global, and motion editing are recommended. In addition, the number of evaluated samples, metric definitions, and statistical variability should be reported more clearly.

**Response 5.7:**

# Response to Reviewer 6

Reviewer 3: The manuscript presents UELVE, which uses an LLM to select local or global and zero-shot or one-shot configurations, arranges video frames into grids for image-diffusion processing, and slides these grids between diffusion steps to promote temporal consistency. The framework also distributes grid batches across multiple GPUs and is evaluated against nine baselines using four automatic quality metrics and runtime measurements for sequences of up to 200 frames. The authors claim consistently superior editing quality, temporal coherence, and efficiency. These claims exceed the evidence because the experiments do not substantiate long-video operation, isolate the LLM contribution, or provide a sufficiently specified and statistically supported evaluation.

**Comment 6.1**

The technical advance over the cited Grid Diffusion method is not isolated, and the manuscript does not define tile resolution, final grid resolution, padding, frame ordering, boundary handling, or the exact application of sliding during inversion and denoising. Provide executable pseudocode and a matched Grid Diffusion baseline, then quantify the effects of sliding and grid size over the complete evaluation set.

**Response 6.1:**

**Frame ordering.** Two alternating grid layouts are adopted. At even diffusion timesteps ($\text{Layout}_1$), frames are packed chronologically into consecutive grids according to their original temporal order. At odd timesteps ($\text{Layout}_2$), all grids slide forward by $n$ columns. As a result, individual frames are re-distributed and may move into neighbouring grids across timesteps. This frame-reordering logic is applied synchronously during both DDIM inversion and denoising.

**Boundary handling.** For sequence boundaries introduced by grid sliding, partial grids are allowed and empty slots are filled following our tail-padding rule: replicated boundary frames participate in attention computation only as context and are discarded before decoding, so the output retains the original number $K$ of video frames. For inter-grid boundaries, the sliding mechanism breaks the isolation between independent grids of original Grid Diffusion: frames originating from different grids are grouped together in alternate timesteps, enabling cross-grid self-attention for long-range temporal consistency.

**Comment 6.2**

The LLM component is neither reproducible nor validated, as the exact DeepSeek model, prompts, decoding settings, output schema, taxonomy, and failure-handling procedure are absent. Report routing accuracy and a confusion matrix for all evaluated edit pairs, comparing the LLM with a deterministic rule-based selector and a manual oracle.

**Response 6.2:**

**Comment 6.3**

The central long-video claim is unsupported because quantitative evaluation stops at 200 frames, corresponding to approximately 7-10 seconds at the stated frame rates, although the described datasets contain clips lasting one to three minutes. Evaluate complete clips on one and two GPUs while reporting peak GPU and host memory, swapping overhead, runtime, communication volume, and failures, or restrict the claims to sequences of at most 200 frames.

**Response 6.3:**

**Comment 6.4**

The quantitative results cannot be assessed reliably because the numbers of videos and edit pairs, task-wise composition, frame-sampling procedure, background masks, aggregation method, random seeds, dispersion, and statistical testing are not reported. Provide a complete evaluation manifest and per-task results with sample sizes, uncertainty estimates, and paired confidence intervals or significance tests for the claimed improvements.

**Response 6.4:**

**Comment 6.5**

The comparison conflates editing categories and methods with different capabilities while UELVE adaptively selects masking and fine-tuning, and disabling supported negative prompts does not establish equitable conditions. Conduct category-specific comparisons using matched backbones, resolutions, prompts, control inputs, supported options, and tuning budgets, supplemented by a blinded human assessment of editing fidelity and temporal artefacts.

**Response 6.5:**

**Comment 6.6**

CLIP-T is described as an average distance although higher values are marked as preferable. State the exact formula and model checkpoint, and use "similarity" or reverse the reported direction as appropriate.

**Response 6.6:**

**Comment 6.7**

Correct recurrent language and typographical errors, including "an Large", "gird", "GUP", and "mehtod", and standardise forms such as CLIP-T, Text-to-Image, and long-form.

**Response 6.7:**

**Comment 6.8**

Replace subjective caption terminology such as "overall beauty" with the measured property and make each qualitative caption identify the prompt, configuration, grid size, and selected frame indice

**Response 6.8:**